Design and Control of The Farmer Robotic Arm for Automated Seed Planting

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LITERATURE REVIEW

The rapid growth of agricultural automation has led to the increasing use of robotic systems to address challenges such as labor shortages, precision planting, and efficiency in large-scale farming. Recent advances in robotic arms, machine vision, control algorithms, and artificial intelligence have made it possible to design agricultural robots capable of performing complex tasks such as planting, harvesting, monitoring, and crop handling with high accuracy.

This section presents a review of five recent scientific articles (2020–2025) that focus on the application of robotic systems in agriculture. Each article is analyzed in terms of its objectives, methodologies, and key contributions, with emphasis on the modeling, control, and real-world implementation of robotic arms or related agricultural robots. The purpose of this review is to provide an overview of the current state of agricultural robotics and highlight the research gaps and opportunities that are directly relevant to the design and control of the proposed **Farmer Robotic Arm for Automated Seed Planting**.

I. ROBOTS IN AGRICULTURE: PROSPECTS, IMPACTS, ETHICS, AND POLICY

This paper reviews the growing role of robotics in agriculture, emphasizing how automation addresses challenges such as labor shortages, efficiency demands, and the need for sustainable production. Agricultural robots are being developed for tasks like planting, weeding, harvesting, monitoring, and spraying, integrating sensors, machine vision, and artificial intelligence to achieve precision farming. The authors highlight the benefits of these technologies, including improved productivity, reduced resource usage, and enhanced crop quality. However, they also discuss challenges such as high costs, limited adaptability in outdoor environments, and the need for standardization. Ethical and social concerns are raised, particularly regarding job displacement of farmworkers and the digital divide between large and small farms. The paper concludes that agricultural robotics has strong potential to transform global food systems, but success depends on careful consideration of technological, economic, and policy factors.

II. AN OVERVIEW OF COOPERATIVE ROBOTICS IN AGRICULTURETS IN AGRICULTURE

The research shows how cooperation between robots is shaping the future of agriculture. Instead of working alone, robots are now designed to collaborate with humans or other machines to complete farming tasks more effectively. The review divides cooperative systems into five main groups: human–robot teams, multiple UAVs (drones), multiple UGVs (ground robots), hybrid UAV–UGV systems, and cooperative robotic arms. Each group has specific applications, such as drones for spraying and monitoring, ground robots for weeding and transport, or robotic arms for planting and harvesting. While drones and ground robots are already advancing quickly, cooperative manipulators remain less studied but could become key for precise operations like fruit picking or seed planting. The paper points out the benefits of cooperation—greater efficiency, flexibility, and scalability—but also emphasizes technical hurdles such as synchronization, communication, and adapting to outdoor farm conditions. The authors recommend further focus on hybrid systems and robotic arms.

III. LOCALIZATION AND MAPPING FOR ROBOTS IN AGRICULTURE AND FORESTRY

HERE IS A SURVEY REVIEWS LOCALIZATION AND MAPPING TECHNIQUES TAILORED FOR ROBOTIC SYSTEMS OPERATING IN AGRICULTURAL AND FOREST ENVIRONMENTS. GIVEN THE CHALLENGES OF GNSS SIGNAL LOSS OR INACCURACY IN DENSE FOLIAGE OR REMOTE FIELDS, THE AUTHORS EMPHASIZE THE NEED FOR GNSS-INDEPENDENT METHODS LIKE SLAM

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(Simultaneous Localization and Mapping) and visual odometry. They categorize techniques into filter-based (EKF, particle filter), optimization / graph-based SLAM, and VO methods. The survey also discusses mapping approaches (metric, topological, semantic), data association challenges, and drift correction (loop closures). Many existing works focus on 2D localization; true 3D pose estimation is rare. The study also catalogs datasets used in agriculture/forestry and analyzes methods by accuracy, scalability, and availability. The authors conclude that the field is still emerging: advanced mapping methods (semantic / topological) are underdeveloped, real-time 3D localization in unstructured environments remains a bottleneck, and future work should target robust, scalable, and weather-resilient approaches.

IV. A Review of Robots, Perception, and Tasks in Precision

With an emphasis on perception systems, robotic platforms, and agricultural tasks, this review article examines the state of robotics in precision agriculture today. The authors classify and talk about sensor technologies (like vision, LiDAR, and multispectral imaging), highlighting how they help with environmental sensing, crop health evaluation, and field monitoring. They then examine robotic solutions, especially small land-based robots, for tasks like harvesting, weeding, mapping, and soil analysis. According to the review, there is a growing trend of robotizing other tasks in agriculture, but the majority of robotic systems in use today are mainly employed for monitoring rather than manipulation or actuation. In order to illustrate the practical difficulties in uneven terrain, the authors also provide a case study of the Agri.Q rover, which was deployed in vineyards. In their conclusion, they point out several trends, including growing autonomy, perception integration, and the possibility of increasingly sophisticated functions (like robotic actuation) in agricultural robots of the future.

V. Exploiting the Internet Resources for Autonomous Robots in Agriculture

The architecture presented in this paper improves autonomous agricultural robots by utilizing Internet resources such as cloud computing, IoT, and data analytics. In order to divide computing between the robot's onboard hardware and internet services, the authors suggest combining the Robot Operating System (ROS) with a cloud-based framework (FIWARE). Task sharing, data storage, scalability, and remote monitoring are all supported by this design. A weeding robot that uses laser-based weed removal is used to illustrate this idea; time-sensitive control remains local, while the cloud communicates with the robot for tasks requiring a lot of computation (like image analytics). Metrics including latency, processing time, bandwidth, and response to remote requests are the main focus of the assessment. According to the results, hybrid computation—local and cloud—can increase capability without putting too much strain on the robot's hardware. The study highlights how Internet-enabled robotics can increase agricultural functionality, but it also poses questions regarding security, response times, and connectivity dependability.

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